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Optical soot characterization using two-color laser-induced incandescence (2C-LII) in the soot growth region of a premixed flat flame

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Abstract

The soot growth region of a premixed one-dimensional ethylene/air flame has been investigated using two-color laser-induced incandescence (2C-LII) with focus on optical soot properties. From the 2C-LII technique, primary soot particle sizes were deduced together with the temperature from pyrometry of the laser-heated particles, while the gas temperatures were obtained from pure rotational CARS nitrogen thermometry. Soot particle sizes were also measured from thermophoretically sampled soot particles analyzed by transmission electron microscopy (TEM), and the results showed growing isolated primary soot particles up to a height of 10 mm, after which strong soot aggregation occurred and the increase in soot primary particle size ceased. The measured data was used in the evaluation of the soot absorption function, $E(m)$, and a strong increase by a factor of two was observed from the lowest heights to the highest for assumed constant values of soot density and specific heat at all heights. By comparing the 2C-LII particle sizes with the sizes obtained from TEM, differences were observed. Part of the explanation is that the 2C-LII evaluation did not take aggregation into account, but it is additionally speculated that the thermal accommodation coefficient decreases with height above burner. These results are interesting in view of morphological and composition changes of the soot during the growth process and have implications for the use of 2C-LII as a soot diagnostic technique.
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Keywords: Laser-induced incandescence; Soot formation; Premixed flat flames; Soot optical properties; Soot absorption function

1. Introduction

Although the understanding of soot formation has grown considerably during the last decades, information is still lacking about the details of these processes. A number of mechanisms have been suggested to be the dominating routes to soot formation, see [1] and references therein, and because of the variety of conditions regarding, for example, fuel, flame type, and temperature, different mechanisms may dominate in different flame operation regimes. An increased knowledge of the fundamental underlying processes is essential in the design of combustion devices with reduced or even zero emissions of soot.

The last decade a lot of effort has been made to improve the understanding of soot formation in

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premixed flat flames. Wang and co-workers have in several papers combined a large number of mainly sampling methods [2], and the late D'Alessio and co-workers have mainly focused on optical methods [3] to improve the knowledge on early soot particle inception and growth. Whereas soot formation in a non-premixed flame has no clear time history in most spatial locations, in a premixed flat flame soot formation is more easily characterized in different zones and can be followed in time from the initial particle inception, via surface growth and coagulation, to aggregation and agglomeration. The newly nucleated particles are much more reactive than the aged particles, and the fast initial surface growth is evidenced by primary particle nuclei identified in the interior of aged primary particles with sizes of tens of nanometers [4]. Moreover, the C/H ratio in the nucleated particles is much lower, C/H = 2–3, than for aged soot particles of several tens of nanometers, C/H = 7–8 [5].

An important issue when using optical methods for soot particle measurements is to what extent the differences in morphology and composition influence the optical properties of the soot particles. The optical properties are often described in terms of the complex refractive index, $m = n - ik$. A variety of values of m for soot have been derived, as discussed in [6]. Other representations of soot optical properties are the absorption function, $E(m) = -Im[(m^2 - 1)/(m^2 + 2)]$, and the dimensionless extinction coefficient K_e, which equals $6\pi E(m)$ if the Rayleigh approximation is valid [7]. Numerous investigations have been performed to establish the dependence of optical soot properties on different parameters [7–9], for example the C/H ratio of the soot [10], and the type of fuel [9]. Some studies have dealt with *in situ* flame measurements, where the complex refractive index of soot formed in premixed flat flames has been found to change with position in the flame [11,12], but it was not clear whether the changes with respect to position were due to the differences in chemical composition, the temperature and/or the different degree of agglomeration [8]. The dependence of soot optical properties on soot morphology is thus not fully understood. Part of the reason is the inherent complex nature of soot but may also be related to the experimental methods and interpretation of data. For *ex situ* measurements made on sampled soot particles the properties may differ from those in flames. Contrary, *in situ* optical measurements such as elastic light scattering and extinction are inherently non-intrusive as long as the incident light intensity is low enough to not significantly heat the particles. However, a majority of the optical flame measurements in the past have dealt with extinction measurements at relatively short wavelengths where gas-phase molecules such as polyaromatic hydrocarbons absorb sufficiently to significantly influence the evaluation of the soot extinction. In [13] it was demonstrated that even as long wavelength as 633 nm gave substantial absorption from gas-phase species in premixed flat ethylene/air flames, and that the Nd:YAG laser wavelength at 1064 nm was a preferred alternative for adequate extinction measurements. Because of the mentioned uncertainties in many investigations, new investigations on the potential variation in optical properties with age and morphology are motivated. Scattering and extinction measurements at 1064 nm is one possibility to contribute to such information, but a combination of the fact that both processes are weaker at these longer wavelengths and that the detectors are much more inefficient makes this approach more challenging for accurate measurements of $E(m)$.

A soot diagnostic technique, which has received strongly growing interest, is laser-induced incandescence (LII) [14–16]. The technique can be used for measurements of soot volume fraction as well as for soot particle size in a variety of combustion environments, among those premixed flat flames [17–19]. LII is based on rapid heating of soot particles by a nanosecond laser pulse, and the increased thermal emission from the heated particles is detected. In low-fluence LII the soot particles are heated to temperatures of ~3500 K, from which the particles cool back to the flame temperature due to heat conduction to the surrounding gas. The LII signal decay rate is dependent on the particle surface-to-volume ratio making it possible to extract particle sizes using a time-resolved approach, such as two-color laser-induced incandescence (2C-LII) [20], together with a theoretical model for the technique. In the 2C-LII approach the time-resolved LII signal is measured at two different wavelengths, which improves the accuracy since the two signals can be used to estimate the maximum particle temperature by pyrometry, hence avoiding uncertainties in $E(m)$ otherwise needed in the modeling of the laser pulse absorption.

LII can be used not only to extract soot volume fraction and particle size, but also to infer $E(m)$. Therssen et al. measured the LII signals generated using both the 532 and 1064 nm beams from a Nd:YAG laser in order to derive the ratio of $E(m)$ at the two wavelengths [21]. Also the 2C-LII approach can be used to derive $E(m)$, as first demonstrated by Snelling et al. in an ethylene/air diffusion flame [22]. The dual-wavelength detection scheme makes it possible to determine the temperature difference between the heated and non-heated particles, information that together with an accurate model for the laser pulse absorption may yield the absolute value of $E(m)$.

In this work we have investigated a premixed flat ethylene/air flame at different heights, where changes in composition and morphology can be expected. Two-color LII has been applied

together with other diagnostic techniques: transmission electron microscopy (TEM) combined with thermophoretic sampling for soot particle sizing, and rotational CARS for measurements of gas temperature. From these data, an attempt has been made to draw conclusions on the change of $E(m)$ and the primary particle size in the soot growth region.

2. LII theory and methodology

2.1. LII theory

The model for LII used in this study is described in detail elsewhere [23] and only a brief description is given here. The model describes the energy exchange between the laser pulse and the soot particles and also the subsequent cooling phase after the end of the pulse. The rates of heat and mass transfer are modeled in two time-dependent differential equations, the first being the energy conservation equation and the second the mass conservation equation. As measurements in this work are carried out at low laser fluence, the energy loss due to sublimation of soot matter is assumed to be negligible and the mass conservation equation can be omitted. The energy balance equation, neglecting the relatively small (~1%) contribution from radiation losses [24], can then be written as

$$\underbrace{\frac{\pi^2 d_p^3 E(m) F}{\lambda} g(t)}_{\text{Absorption}} - \underbrace{\frac{1}{8}\alpha_T \pi d_p^2 p \sqrt{\frac{8 k_B T_g}{\pi m_g}} \frac{\gamma+1}{\gamma-1}\left(\frac{T_p}{T_g}-1\right)}_{\text{Heat conduction}} = \underbrace{\frac{\pi}{6} d_p^3 \rho_s c_s \frac{dT_p}{dt}}_{\text{Internal energy}}, \tag{1}$$

where d_p is the primary particle diameter, F the laser fluence, $g(t)$ the temporal distribution of laser energy, λ the laser wavelength, α_T the thermal accommodation coefficient, p the gas pressure, k_B the Boltzmann constant, m_g the average molecular weight of the gas molecules, γ the heat capacity ratio of the gas, T_p and T_g the soot particle and gas temperatures, and ρ_s and c_s the density and specific heat of soot, respectively. In Eq. (1) the heat conduction model has been given explicitly for the free molecular regime to clarify the dependence of this sub-mechanism on various parameters, whereas in this work the more advanced Fuchs heat conduction model has been used [23]. However, the Fuchs model is more complex and cannot be given analytically in closed form. The particle size dependence of the LII signal originates from the fact that the heat conduction loss mechanism is dependent on the soot surface area, whereas the internal energy is volume-dependent.

After solving the energy balance equation for the variables T_p and d_p, the LII signal can be determined using the Planck radiation law combined with the emissivity for soot according to

$$S_{\text{LII}} \propto \pi d_p^2 \frac{4\pi d_p E(m)}{\lambda_{det}} \times \frac{2\pi h c^2}{\lambda_{det}^5}\left(\frac{1}{e^{hc/\lambda_{det} k_B T}-1} - \frac{1}{e^{hc/\lambda_{det} k_B T_g}-1}\right), \tag{2}$$

where h is the Planck constant, and c the speed of light. The detection wavelength, λ_{det}, in Eq. (2) is assumed to be single, which is an assumption that can be made since the LII signal is measured using spectrally narrow-band filters, as previously discussed in [25].

2.2. Primary particle size data evaluation

By measuring the LII signal at two different wavelengths, the particle temperature can be calculated using the expression [20]

$$T_p = \frac{hc}{k_B}\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)\left[\ln\left(\frac{S_1}{S_2}\frac{C_2}{C_1}\frac{\lambda_1^6}{\lambda_2^6}\right)\right]^{-1}, \tag{3}$$

where λ_i is the detection wavelength, S_i the LII signal and C_i the calibration constant for each of the two wavelengths, parameters that are experimentally determined. Eq. (3) is based on the assumption of uniform heating, i.e. a situation where all soot particles in the measurement volume reach the same maximum temperature during the pulse. In the present study this was assured by using a top-hat spatial profile of the laser.

The approach for extracting particle sizes in this work is based on adjusting the absolute strength of the absorption term in the model in order to obtain a modeled particle temperature, T_0, equal to the one obtained experimentally using Eq. (2) at the same time position, t_0, which is set at the start of data evaluation. To be sure no residual laser pulse energy is affecting the decay, t_0 was chosen 50 ns after the peak of the LII signal. Particle size evaluation in this work was performed assuming a monodisperse size distribution, because attempts to fit polydisperse distribution functions to the experimental signals were not successful at all spatial positions. It should be noted that it is generally challenging to fit polydisperse distributions to LII signals due to the fitting problem being ill-posed [26]. In order to compare the evaluated monodisperse LII sizes with the polydisperse distributions obtained from our TEM analysis the following approach was made. A lognormal size distribution function was fitted to the size data from TEM, and the corresponding LII signal response from this distribution was calculated using the LII model. In a next step

curve fitting was carried out and the best match among model predictions for monodisperse size distributions was found. This best fit size is referred to as the LII monodisperse equivalent size and is always larger than the mean of the polydisperse TEM distribution, as a result from the LII technique biasing the average results towards larger sizes [25,27].

A crucial parameter in the model, which is associated with quite large uncertainty and is especially important when evaluating particle sizes, is the thermal accommodation coefficient α_T, which determines the heat transfer efficiency between an impinging gas molecule and the particle surface [28]. This parameter was determined experimentally by comparing LII signal predictions for the monodisperse equivalent LII size obtained from TEM with the experimentally obtained LII signals from the flame. In a previous study, where only three positions were studied [25], attempts were made to derive individual values of α_T in a similar flame, and it was found that α_T appeared to decrease with increasing height above burner. The much more extensive TEM data analysis carried out for this work, covering heights from 6 to 17 mm, have shed some doubt with regard to this approach as it is subjective to the uncertainties present in both the LII and TEM analysis, especially so for the positions with high level of aggregation. The decision was therefore to initially treat α_T as constant as function of height and find a best fit at 9 mm where particle aggregation is marginal, and hence TEM data are more trustworthy.

2.3. Evaluation of E(m) at 1064 nm

The absorption function $E(m)$ appears in the absorption term in Eq. (1). By accurately measuring the gas temperature, T_g, and the maximum soot particle temperature, $T_{p,\max}$, this difference can be used to calculate $E(m)$ using the model for LII given in Section 2.1. The methodology is similar to the one outlined in Section 2.2, by adjusting the strength of the absorption term until a best fit is found between predicted and measured soot particle temperature. However, there is one fundamental difference: When evaluating particle sizes, as described in Section 2.2, no physical interpretation is made from the strength of the absorption term. The strength is adjusted to obtain a start temperature for the evaluation starting 50 ns after the peak, and the only physical quantity inferred is the particle size. When evaluating $E(m)$, this particle size is used in the model, but its influence on the results is minimized by evaluating $E(m)$ at a time position where the LII signal has its peak. This scheme also ensures maximum signal-to-noise ratio, which is especially important at the lower heights in the flame. In order to calculate $E(m)$, i.e. infer a physical quantity from the strength of the absorption term, it is also necessary to accurately measure the laser pulse energy and temporal and spatial profile of the laser.

3. Experimental procedures

The measurements were performed in a premixed flat ethylene/air flame with a stoichiometry of $\Phi = 2.1$ and a total flow of 10 l/min (at 0 °C and 1 atm). The flame burned on a so-called McKenna burner (Holthuis Inc. [29]), which has a sintered bronze plug with a diameter of 60 mm. An outer co-flow of nitrogen shielded the flame from room air entrainment. A flame stabilizer of stainless steel (cylindrical with a diameter of 60 mm) was mounted with its lower edge at a distance of 21 mm above the burner surface.

Gas temperatures were measured using rotational coherent anti-Stokes Raman spectroscopy (CARS) [30]. In this work, a dual-broad-band pure rotational CARS approach was used, which has been utilized for a vast number of projects where temperature information has been needed, also previously in sooting flames [31]. The general uncertainty in rotational CARS nitrogen thermometry is often considered to be up to 2–3% of the absolute temperature. More information about the experimental setup and the details of the technique can be found elsewhere [31].

The experimental setup for LII has been described in detail elsewhere [25] and is only briefly outlined here. The system is schematically shown in Fig. 1. The beam from a Q-switched Nd:YAG laser at 1064 nm was expanded and a central uniform rectangular-shaped portion of the beam was relay imaged into the centre of the flame where it provided a 1 mm high and 4.8 mm wide top-hat beam. The online beam monitoring was carried out before the flame in order to measure the beam properties unaffected by the flame. The LII signal was imaged onto a slit selecting a width of 10 mm along the optical axis of the laser beam and was detected at

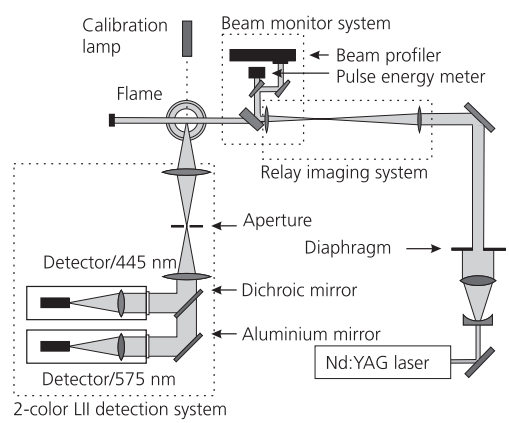

Fig. 1. The experimental setup for 2C-LII.

445 and 575 nm using two separate photomultiplier tubes (PMTs). The dimensions of the beam determined the spatial resolution of the measurement system in the height and depth dimension whereas the width of the aperture determined the resolution along the beam axis. The resolution as function of height is important as the soot formation process occurs along this axis and a higher resolution (i.e. a laser beam height <1 mm) would be beneficial. However, the limited signal-to-nose ratio at the lower heights above burner put constraints in this respect and the decision was made to accept some averaging. The laser fluence was kept at 0.13 J/cm^2 and the temporal distribution of laser energy was monitored using a 10 GHz photodiode (not shown in Fig. 1). Both the PMT signals and the signals from the photodiode were detected and stored on a digital 1 GHz oscilloscope. For each height 1000 single shots were recorded. The detectors were calibrated using a broad-band calibration lamp. The peak-normalized shot-averaged LII signals for a selected part of the data set is shown in Fig. 2. It can be clearly seen how the decay rate decreases with height above burner up to around 13 mm, which mainly can be related to an increase in particle size.

Thermophoretic sampling was achieved using a pneumatically driven soot sampling probe on which thin carbon-coated copper grids were attached. At each measurement position two measurements were carried out, and the residence time was ~300 ms. The sampled soot was later analyzed using a 100 kV transmission electron microscope and images were recorded at two separate positions using the magnification of 39,000 for heights 6–8 mm and 28,800 for higher positions. The negatives were scanned at 1200 dpi and manually analyzed using the freeware ImageJ [32].

4. Results

The particle sizes from the TEM analysis are shown in Fig. 3 together with the evaluated particle sizes from LII data according to the methodology outlined in Section 2.2. The TEM analysis resulted in two curves, both the arithmetic mean of the primary particle size distributions at each position, $\langle d_{p,\mathrm{TEM}}\rangle$, and the monodisperse equivalent size, $d_{p,\mathrm{me,TEM}}$. The latter parameter should be interpreted as an average size evaluated from an LII experiment when measuring a polydisperse distribution described by the TEM data. The monodisperse equivalent size is always larger than the arithmetic mean as previously discussed in Section 2.2. It can be observed that the evaluated TEM primary particle sizes increase fast up to a height of 10 mm after which they become almost constant. This is also the height where the first aggregates become frequent in the TEM images. Up to 9 mm, TEM images show dominantly isolated particles, which is exemplified in the insert in Fig. 3 at 6 mm. At a height of 10 mm the TEM images show some small aggregates, and at higher heights the primary particle size is marginally changed while the aggregation strongly increases, as shown in the inserts in Fig. 3.

By comparing the primary particle size obtained from LII, $d_{p,\mathrm{me,LII}}$, with the monodisperse equivalent obtained from the TEM analysis, $d_{p,\mathrm{me,TEM}}$, at 9 mm above burner, an α_T of 0.48 was obtained. Using this value, $d_{p,\mathrm{me,LII}}$ was evaluated at all heights, and the results are shown in Fig. 3 as well. Interestingly, the evaluated LII particle sizes increase more strongly as function of

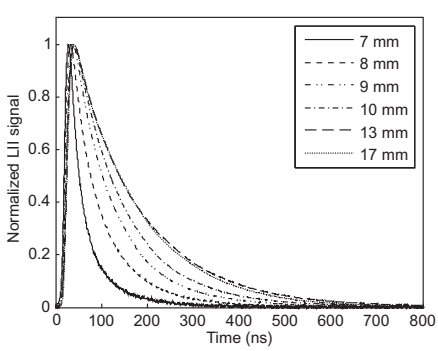

Fig. 2. Time-resolved peak-normalized LII signals for the flat ethylene/air flame. The signals are averaged over 1000 shots.

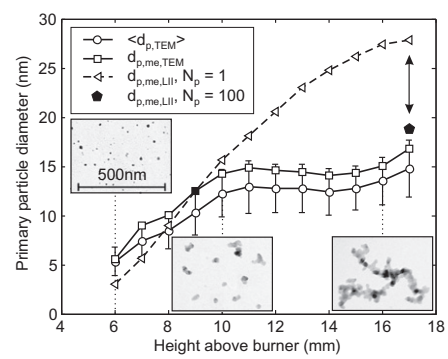

Fig. 3. The primary particle size as function of height in the flat ethylene/air flame. The circles represent the arithmetic mean of the polydisperse distribution functions obtained from TEM, $\langle d_{p,\mathrm{TEM}}\rangle$, where the error bars show the standard deviation. Also shown is the monodisperse equivalent size, $d_{p,\mathrm{me,TEM}}$, derived from the TEM data. The sizes evaluated from the experimentally obtained LII signals (Fig. 2) are given as $d_{p,\mathrm{me,LII}}$. These were evaluated using a constant thermal accommodation coefficient ($\alpha_T = 0.48$) derived by comparing with $d_{p,\mathrm{me,TEM}}$ at 9 mm height above burner, as marked by the filled squared symbol. The black pentagon marks the evaluated results assuming aggregates of 100 primary particles ($N_p = 100$). The inserts show examples from the TEM images, all with the same magnification.

height than the TEM sizes. An important factor in this respect is aggregation of primary soot particles, which influence the decay rate of LII signals as the heat conduction is reduced due to the shielding effect [33]. Taking aggregation into account would decrease the evaluated particle sizes from LII at heights above 10 mm, a height at which aggregation was found to start and become stronger at increasing height. The TEM data available for the present work is not sufficient to determine aggregate size distributions and hence no attempt was made to take aggregation into account for all positions. However, to give an example of the magnitude of the aggregation effect, the LII results at 17 mm were evaluated assuming aggregates with the number of particles per aggregate, $N_p = 100$ (black pentagon in Fig. 3) using the aggregation sub-model described in Liu et al. [33]. As can be seen, the influence on evaluated primary particle size is substantial.

In Fig. 4 the gas temperature evaluated by CARS, $T_{g,\mathrm{CARS}}$, is shown together with the maximum particle temperature, $T_{p,\max}$, evaluated from the LII signals following the approach in Section 2.2. At low heights the gas temperature drops slightly for increasing heights because of radiation losses, and at heights above ~13 mm additionally from the cooling by the flame stabilizer acting as a heat sink. $T_{p,\max}$ increases steadily with height above burner despite a constant laser pulse energy passing the flame. The temperature difference, $T_{p,\max} - T_{g,\mathrm{CARS}}$, increases strongly with height, and this may be interpreted as an increase in the value of the absorption function $E(m)$ at 1064 nm as shown in Fig. 4.

5. Discussion

The results in Fig. 4 show how the absorption function, $E(m)$, varies from values around 0.21 at the lowest measured position in the flame, increases with height and approaches a value of 0.45 at the highest position. These values are within the range typically reported for flame soot [15]. In the work by Snelling, where the present procedure for using two-color LII for determination of $E(m)$ was first introduced [22], $E(m)$ was evaluated to 0.395 in a specific flame location where the soot particles can be considered to be old, less reactive, and aggregated. In our flame, we consider that the soot particles at flame heights 16–17 mm have optical properties closest resembling those of [22], and Fig. 4 shows that the evaluated $E(m)$ approaches an almost constant value of 0.45 at these heights. When using the LII model parameters reported in [22] ($\rho_s = 1.9$ g/cm^3, $c_s = 2100$ J/kg K and $\alpha_T = 0.37$) in the evaluation of our LII data at 17 mm above the burner, a value of $E(m)$ of 0.40 was reached, thus in close agreement with 0.395. Although agreement was reached at a height of 17 mm in the flame, it is important to note that the behavior of $E(m)$ shown in Fig. 4 is evaluated using certain assumptions. The question arises if the relatively strong dependence with height shown in Fig. 4 is really due to a change in refractive index, or some other factors. The values extracted with the present methodology will be dependent on uncertainties both in experimental data, theoretical models and input parameters for these models. We estimate the uncertainties in the measurements of gas temperature and soot particle temperature to give an uncertainty of up to 5% in $E(m)$. Also, both the way absorption and internal energy of soot particles are modeled must be taken into consideration. The absorption term in Eq. (1) is valid for the Rayleigh regime (and also for fractal aggregates following the Rayleigh–Debye–Gans approximation [15]), and as the present study is carried out using a 1064 nm laser beam with primary particles diameters less than 20 nm, the risk of introducing considerable uncertainty in this model is believed to be small. Indeed, following the results from Liu et al. [34] the error should be less than 10% for all particles encountered in the TEM analysis.

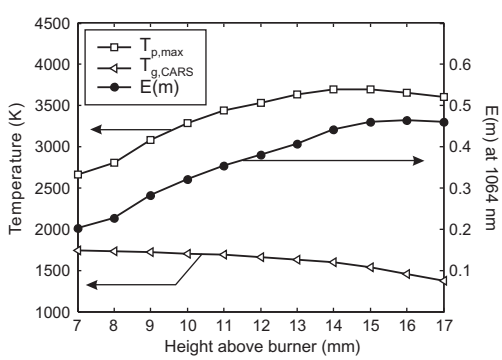

Fig. 4. Evaluated maximum soot particle temperatures, $T_{p,\max}$, flame temperatures, $T_{g,\mathrm{CARS}}$, and the absorption function, $E(m)$, as function of height above burner.

Another important fact is that this evaluation of $E(m)$ is based on assumptions of soot particle density, ρ_s, and specific heat, c_s, quantities that are uncertain in most practical situations. In this work the temperature-dependent expressions for ρ_s and c_s given by Michelsen were used [35]. These values are for graphite and may not be applicable to the smallest particles in the growth region of the premixed flame in this study. For example, Dobbins has reported that the density of the nucleated particles is substantially lower (~1.2 g/cm^3) [36] than those typically used for graphite (~1.9 g/cm^3) and recently Totton et al. [37] reported similar values for nascent soot particles. Interestingly, using lower values of the soot density in our evaluation would result in even lower values of $E(m)$, and considering the density to

increase with height above burner, the inferred variation of $E(m)$ with height would be even more pronounced. Moreover, the trend of increasing $E(m)$ with height has been seen before. Studies using combined elastic scattering/extinction and dynamic light scattering have shown some increase in both the real and imaginary part of the complex refractive index of soot, leading to an increase in $E(m)$ of about 20%, from low to high heights in the flame [11,12].

Considering the results presented in Fig. 3, it is evident how the primary particle sizes evaluated from the TEM data and the LII signals differ quite substantially. As previously mentioned the level of aggregation was not accounted for in the evaluation, and results using the aggregation model for the point at 17 mm clearly demonstrated that the effect can be large. Indeed, TEM data and model results indicate a systematic error of ~50% due to the decreased heat conduction at 17 mm. Due to the relatively strong effect of aggregation on the results, we expect the main uncertainty for particle sizing at heights above 10 mm where aggregation is substantial is because this effect was not taken into account in the LII model. Additional uncertainty exists in the TEM data where it becomes increasingly hard to identify spherical primary constituents from the highly aggregated particles high in the flame. Also at the lower heights in the flame, the LII particle sizes increase much faster than the TEM sizes with increasing height, despite the presence of isolated particles only. We speculate that the main uncertainty in particle sizing in this flame region is because of a limited knowledge of α_T, ρ_s and c_s. In the early stage of growth, the soot particles undergo fast changes in their chemical and physical properties, and it is not farfetched to expect changes in these parameters as function of height. If an individual α_T had been fitted at each height in the flame while keeping ρ_s and c_s constant, the result had been an initially very high value on α_T that drops for increasing heights, and such a trend was indicated in a previous study from our group [25]. It becomes difficult to estimate the uncertainty in the measurements as a result of changes in the mentioned parameters as such knowledge is limited. To the authors' knowledge no literature data is available on the variation of the three parameters as function of time in a soot formation process (or in our case as function of height in the flat flame) and future research will be needed to fill the knowledge gap on the properties of the young particles in the nucleation and growth region.

6. Conclusions

The aim of this work has been to get new insights into soot characteristics of a premixed flat flame using laser-induced incandescence (LII) in combination with analysis of thermophoretically sampled soot particles by transmission electron microscopy (TEM) and rotational CARS thermometry. In this study, two-color LII has been used to gain information about the change in optical properties in the soot growth region in this type of flame. The peak soot temperature was evaluated from two-color pyrometry after being heated by a nanosecond pulse at 1064 nm at a specific energy, and estimated temperatures were found to increase strongly with height above burner. The difference between the temperatures of the heated particles and the gas temperature was used to calculate a value of the absorption function $E(m)$ for assumed constant soot density and soot specific heat valid for graphite. As the value of these parameters are expected to deviate from those of graphite low in the studied flame, the evaluated $E(m)$ is here more uncertain, but the observed trend of increasing $E(m)$ with height in evaluated results should still be true. In the comparison of particle sizes estimated from TEM and from 2C-LII, a thermal accommodation coefficient of 0.48 gave agreement in particle size at a height of 9 mm. When using this value at all heights in the flame, the evaluated soot particle sizes using LII was much smaller at lower heights and much larger at higher heights than those analyzed by TEM. This behavior may be described by a decreasing α_T as function of height, but because of uncertainties in the variation of density and specific heat of soot with height, and most specifically the uncertainty in how aggregation affects our LII signals, this result is not conclusive. The observations in this work are very interesting and will be followed by new investigations where complementary diagnostic techniques will be used and a more advanced aggregation sub-model will be integrated in the LII model.

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